

Sumith Chandra Satkuri

AI/ML Engineer

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SUMMARY

AI/ML Engineer focused on making large language models efficient and production-ready — LLM fine-tuning, quantization, and agentic RAG. Samsung R&D intern who cut LLM size 50% and doubled inference speed; independently engineered a full-stack dual-LLM agentic RAG application end-to-end. Now shipping production software under tight deadlines using AI-assisted development workflows. IIT Kharagpur CS graduate and Codeforces Candidate Master with strong algorithmic and systems fundamentals.

SKILLS

- **Languages:** Python, C++, Go, SQL, TypeScript
- **AI/ML:** PyTorch, Transformers, LLMs, LoRA/QLoRA, PTQ/AWQ quantization, DeepSpeed, RAG, Agentic AI
- **Tools & Web:** LangChain, LangGraph, ChromaDB, FastAPI, React, Hugging Face, Git, Linux

EXPERIENCE

Software Engineer — Insurity, Noida, India Jun 2026 – Present

- Modernizing a legacy ColdFusion application into a React + TypeScript front end, improving maintainability
- Translating legacy business logic into typed, component-based React with clean, self-documenting code.
- Delivered time-boxed migration tasks with high correctness by orchestrating Claude Code and Replit in a cross-verification workflow, reviewing both outputs to optimize speed without sacrificing accuracy.

AI/ML Research Intern — Samsung R&D Institute, Bangalore, India May 2025 – Jul 2025

Fine-tuning LLMs Under Resource Constraints

- Engineered PTQ and AWQ quantization pipelines that cut LLM size 50% and doubled inference speed.
- Reduced fine-tuning GPU memory 60% using LoRA and QLoRA parameter-efficient adaptation.
- Enabled 4x longer-sequence training in fixed memory via DeepSpeed ZeRO parallelism and Liger-kernel.
- Shipped an on-device LLM retaining 95%+ task accuracy via quantization, PEFT, and distributed training.

PROJECTS

PaperMind — AI-Powered Agentic RAG Assistant Apr 2026 – May 2026

github.com/sumith2425/PaperMind

- Built an agentic LangGraph router across 3 routes: document RAG, web search, direct answer.
- Engineered a ChromaDB RAG pipeline with cosine-similarity fallback to Tavily web search for page-cited QA.
- Delivered a streaming multi-turn chat UI (FastAPI + SSE, React 19) on a dual-LLM stack (Llama 3.1 8B + 3.3 70B) via Groq.

Defending Deep Learning Models Against Adversarial Attacks — B.Tech Project Aug 2025 – Apr 2026

github.com/sumith2425/Defending-Deep-Learning-Models-against-Adversarial-Attacks

- Restored VideoMAE robust accuracy from 0% to 85%+ under PGD attacks via attack-type-routing defense.
- Held 94%+ clean accuracy across PGD, OFA, C&W attacks via a ResNet-18 flow + Integrated Gradients detector.
- Neutralized adversarial perturbations through adaptive spatial-transformation routing (resizing, cropping).

SQL Query Parser & Optimizer — C, Flex, Bison Jan 2025 - Apr 2025

github.com/sumith2425/SQL_Parser_and_Optimizer

- Built a SQL engine in C that lexes and parses queries into relational algebra trees using Flex and Bison.
- Added pushdown and join-reordering passes with cost-based selection over 25 candidate plans.

EDUCATION

Indian Institute of Technology (IIT) Kharagpur Nov 2022 – May 2026

B.Tech (Hons.), Computer Science and Engineering • **CGPA: 8.74 / 10** • Kharagpur, India

- Coursework: Machine Learning, Deep Learning, Information Retrieval, High Performance Parallel Programming, Operating Systems, Computer Networks, DBMS.

ACHIEVEMENTS

- Codeforces Candidate Master — rating 1923 (handle: Sumith_chandra).
- JEE Advanced 2022 — All India Rank 883
- JEE Main 2022 — All India Rank 1329.
- National Talent Search Examination (NTSE) Scholar, NCERT.